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MJ (Ma

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FSK116 Part A of Semester Test 2 (Test)

Current Score

QUESTION

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2

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9

Due Date

TUE, JUN 8, 2021
2:30 PM GMT+2

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Instructions

*Answer all questions. *Give your final answers to the correct number of significant figures. *Your final answers for large and small numbers need to be written in scientific notation. (In WebAssign the format for example for representing 2.87×10^{-5} is 2.87 x 10 ^-5). *Insert a space between your numerical answer and the units if units are needed. *Also use the PhysPad/MathsPad for mathematical layouts, such as roots, fractions, superscripts and subscripts etc. *In symbolic format, for example, $2mg$ is $2 \cdot m \cdot g$ where the $\cdot$ is the dot / multiply operator. *Use a laptop computer instead of a smartphone, for, smartphones do not work with the PhysPad/MathsPad. *This is a timed assessment. *You have 60 minutes (1 hour) to complete Part A. ie. deadline is 14h30. *Just before your final submission, print out your now filled-in WebAssign Part A test (there is a "print" tab at the top of the test, it prints to a PDF file - you don't need a physical printer!). *Submit it into the relevant part in the "Semester and Online Tests Folder" on ClickUP. *Do this submission immediately after submitting the WebAssign Part A test. This is what the 15 min gap is for between the end of Part A and the beginning of Part B. *NO late submissions allowed! *You have 2

Assignment Submission & Scoring

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit and change the answer.

Assignment Scoring

Your last submission is your score.

submission opportunities in WebAssign - but these must be completed within the allotted time for this activity.

 Description



1. **DETAILS** **PREVIOUS ANSWERS** **SERPSE10 4.6.OP.026.**

MY NOTES

ASK YOUR TEACHER

A plane is flying to a city **756** km directly north of its initial location. The plane's speed of **168** km/h relative to the air during its flight.

- (a) If the plane flies through a constant headwind blowing south at **57.5** km/h, how much time (in h) will it take to reach the city?



h

- (b) If instead the plane flies through a constant tailwind blowing at **57.5** km/h, how much time (in h) will it take to reach the city?



h

- (c) If instead the plane flies through a constant crosswind blowing east at **57.5** km/h, how much time (in h) will it take to reach the city?



h

2.

DETAILS

PREVIOUS ANSWERS

SERPSE10 5.8.OP.030.

MY NOTES

ASK YOUR TEACHER

An 11.5 g bullet is moving to the right with speed 250 m/s when it hits a target an additional 22.6 cm into the target. What are the magnitude (in N) and direct stopping force acting on the bullet? Assume the stopping force is constant.

magnitude



N

direction



3.

DETAILS

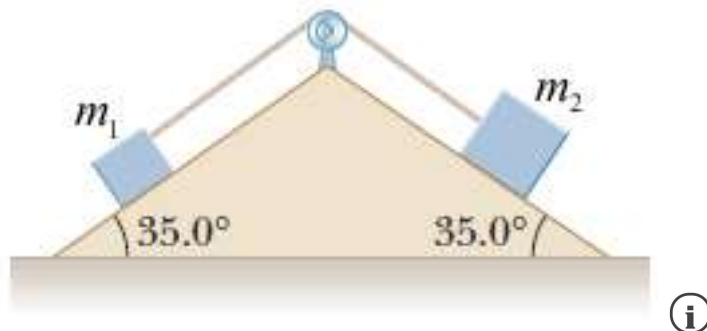
PREVIOUS ANSWERS

SERPSE10 5.8.OP.039.

MY NOTES

ASK YOUR TEACHER

A cord passing over a pulley connects two masses, as shown, where $m_1 = 4.10$ kg and $m_2 = 6.70$ kg. The system accelerates with a magnitude of 0.83 m/s². The coefficient of kinetic friction between the masses and the incline is the same for both masses. The pulley is frictionless, and the cord is massless. (Due to the nature of this problem, do not use rounded intermediate values in your calculations—including answers submitted to WebAssign.)



(a) What is the coefficient of kinetic friction?

 ✖

(b) What is the tension (in N) in the cord?

 ✖ N

4.

DETAILS

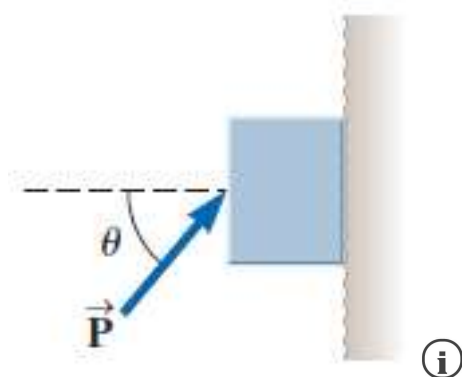
PREVIOUS ANSWERS

SERPSE10 5.8.P.034.

MY NOTES

ASK YOUR TEACHER

A block of mass **2.40** kg is pushed up against a wall by a force $\vec{P}$ that makes a $\theta = 50.0^\circ$ angle with the horizontal as shown below. The coefficient of static friction between the block and the wall is **0.280**.



(a) Determine the possible values for the magnitude of $\vec{P}$ that allow the block to remain stationary. (If there is no maximum, enter NONE in that answer.)

$$|\vec{P}_{\max}| = \boxed{\text{NONE}} \times \text{ N}$$

$$|\vec{P}_{\min}| = \boxed{24.9} \checkmark \text{ N}$$

(b) What happens if $|\vec{P}|$ has a larger value than $|\vec{P}_{\max}|$?

- ☒ The block slides up the wall.
- ☐ The block does not slide along the wall.
- ☐ The block slides down the wall.



What happens if $|\vec{P}|$ has a smaller value than $|\vec{P}_{\min}|$?

- ☐ The block slides up the wall.
- ☐ The block does not slide along the wall.
- ☒ The block slides down the wall.



(c) Repeat parts (a) and (b) assuming the force makes an angle of θ with the horizontal. Determine the possible values for the magnitude of $\vec{P}$ that allow the block to remain stationary. (If there is no maximum, enter ∞ for the maximum, and if there is no minimum, enter blank.)

$$|\vec{P}_{\max}| = \boxed{} \text{ N}$$

$$|\vec{P}_{\min}| = \boxed{} \text{ N}$$

What happens if $|\vec{P}|$ has a larger value than $|\vec{P}_{\max}|$?

- ☐ The block slides up the wall.
- ☒ The block does not slide along the wall.
- ☐ The block slides down the wall.



What happens if $|\vec{P}|$ has a smaller value than $|\vec{P}_{\min}|$?

- ☐ The block slides up the wall.
- ☐ The block does not slide along the wall.
- ☒ The block slides down the wall.



5.

DETAILS

PREVIOUS ANSWERS

SERPSE10 6.2.OP.009.

MY NOTES

ASK YOUR TEACHER

- (a) A stuntman with a mass of 87.5 kg swings across a moat from a rope that is 10.0 m long. At the bottom of the swing the stuntman's speed is 7.35 m/s. The rope's maximum strength is $1,000$ N. Will the stuntman make it across the moat without falling?

☐ Yes☒ No

- (b) **What If?** What is the maximum speed (in m/s) that the stuntman can have at the bottom of the swing on this vine to safely swing across the river?



m/s

6.

DETAILS

SERPSE10 6.4.OP.015.

MY NOTES

ASK YOUR TEACHER

- (a) A sphere made of rubber has a density of 0.920 g/cm^3 and a radius of 8.5 falls through air of density 1.20 kg/m^3 and has a drag coefficient of 0.500 terminal speed (in m/s)?

 m/s

- (b) From what height (in m) would the sphere have to be dropped to reach the height it fell without air resistance?

 m

7.

DETAILS

PREVIOUS ANSWERS

SERPSE10 7.4.P.012.

MY NOTES

ASK YOUR TEACHER

Express the units of the force constant of a spring in SI fundamental units. (Use meter, kg for kilogram, and s for second in your equation.)

 $[k] =$ N/m 

8.

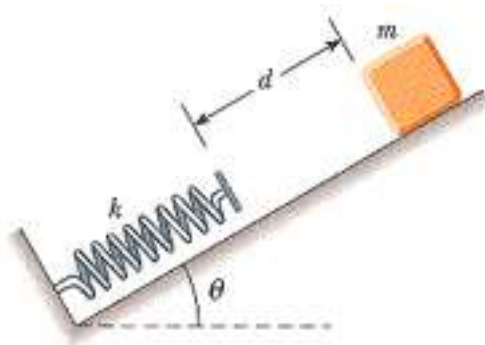
DETAILS

SERPSE10 8.XP.005.

MY NOTES

ASK YOUR TE

An object of mass m starts from rest and slides a distance d down a frictionless angle θ . While sliding, it contacts an unstressed spring of negligible mass as shown in the figure. The object slides an additional distance x as it is brought momentarily to rest by the compression of the spring (of force constant k). Find the initial separation d between the object and spring. (Use any variable or symbol stated above along with the force of gravity: g .)

 $d =$ 

9.

DETAILS

PREVIOUS ANSWERS

SERPSE10 8.XP.008.

MY NOTES

ASK YOUR TEACHER

A single constant force $\mathbf{F} = (4\mathbf{i} + 6\mathbf{j})$ N acts on a 4.08 kg particle.

(a) Calculate the work done by this force if the particle moves from the point having the vector position $\mathbf{r} = (2\mathbf{i} - 3\mathbf{j})$ m.

J Does this result depend on the path?

☒ Yes☐ No

Explain.

(b) What is the speed of the particle at $\mathbf{r}$ if its speed at the origin is 4

m/s

(c) What is the change in the potential energy?

J

Submitted

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